



MANIPAL

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MANIPAL SCHOOL OF INFORMATION SCIENCES

(A Constituent unit of MAHE, Manipal)

Cryptology – 6_Lab

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Write a python code to perform One Time Pad

```
import secrets
import base64

def generate_key(length):
    """Create a random key with the same length as the message."""
    return secrets.token_bytes(length)

def encrypt(message, key):
    """Encrypt the message using XOR."""
    message_bytes = message.encode("utf-8")

    encrypted_bytes = bytes(
        message_byte ^ key_byte
        for message_byte, key_byte in zip(message_bytes, key)
    )

    # Convert encrypted bytes into printable text
    return base64.b64encode(encrypted_bytes).decode("utf-8")

def decrypt(encrypted_message, key):
    """Decrypt the message using XOR."""
    encrypted_bytes = base64.b64decode(encrypted_message)

    decrypted_bytes = bytes(
        encrypted_byte ^ key_byte
        for encrypted_byte, key_byte in zip(encrypted_bytes, key)
    )

    return decrypted_bytes.decode("utf-8")

# Original message
message = "Hello, World!"

# Create a random key with the same length as the message
key = generate_key(len(message.encode("utf-8")))

# Encrypt the message
encrypted_message = encrypt(message, key)
print("Encrypted:", encrypted_message)

# Decrypt the message
decrypted_message = decrypt(encrypted_message, key)
print("Decrypted:", decrypted_message)
```

```
● haris@MacBook-M1 6_Lab % python3 1.py
Encrypted: /KK2C3i8lp1AhKF6Dw==
Decrypted: Hello, World!
```

Write a Python code to break the shift cipher "MCI RWR WH" (Perform a brute-force attack by displaying all 26 possible plaintexts.)

```
def decrypt_shift_cipher(text, shift):
    result = ""

    for character in text:
        if character.isalpha():
            # Convert the character to a number from 0 to 25
            number = ord(character.upper()) - ord('A')

            # Shift the character backward
            decrypted_number = (number - shift) % 26

            # Convert the number back to a letter
            result += chr(decrypted_number + ord('A'))
        else:
            # Keep spaces and punctuation unchanged
            result += character

    return result

ciphertext = "MCI RWR WH"

# Try every possible shift
for shift in range(26):
    decrypted_text = decrypt_shift_cipher(ciphertext, shift)
    print("Shift", shift, ":", decrypted_text)
```

```
haris@MacBook-M1 6_Lab % python3 2.py
Shift 0 : MCI RWR WH
Shift 1 : LBH QVQ VG
Shift 2 : KAG PUP UF
Shift 3 : JZF OTO TE
Shift 4 : IYE NSN SD
Shift 5 : HXD MRM RC
Shift 6 : GWC LQL QB
Shift 7 : FVB KPK PA
Shift 8 : EUA JOJ OZ
Shift 9 : DTZ INI NY
Shift 10 : CSY HMH MX
Shift 11 : BRX GLG LW
Shift 12 : AQW FKF KV
Shift 13 : ZPV EJE JU
Shift 14 : YOU DID IT
Shift 15 : XNT CHC HS
Shift 16 : WMS BGB GR
Shift 17 : VLR AFA FQ
Shift 18 : UKQ ZEZ EP
Shift 19 : TJP YDY DO
Shift 20 : SIO XCX CN
Shift 21 : RHN WBW BM
Shift 22 : QGM VAV AL
Shift 23 : PFL UZU ZK
Shift 24 : OEK TYT YJ
```

Write a Python code to encrypt Rail Fence

```
def rail_fence_encrypt(text, rails):
    # If there is only one rail, no encryption is needed
    if rails == 1:
        return text

    rows = [""] * rails
    row = 0
    direction = 1

    for character in text:
        rows[row] += character

        # Change direction at the top or bottom rail
        if row == 0:
            direction = 1
        elif row == rails - 1:
            direction = -1

        row += direction
```

```
        row += direction

    # Join all the rails together
    return "".join(rows)

message = "HELLO WORLD"
rails = 3

encrypted_message = rail_fence_encrypt(message, rails)

print("Original message:", message)
print("Encrypted message:", encrypted_message)
```

```
● haris@MacBook-M1 6_Lab % python3 3.py
Original message: HELLO WORLD
Encrypted message: HOREL OLLWD
○ haris@MacBook-M1 6_Lab %
```